

Best Time to Buy and Sell Stock

1. Understand the Problem

We have stock prices:

$$[7, 1, 5, 3, 6, 4]$$

We must:

Buy first, then sell later

Our goal is to make the biggest possible profit.

For example:

$$buy = 1, \quad sell = 6$$

so:

$$profit = 6 - 1 = 5$$

The important restriction is:

Buy must happen before sell

2. Start With the Simple Idea

One simple approach is to try every possible pair:

$$buy_i, \quad sell_j$$

where:

$$i < j$$

For every pair:

$$profit = prices[j] - prices[i]$$

Then keep the largest profit.

The problem is that there can be about:

$$O(n^2)$$

pairs.

With $n = 10^5$, this is too slow.

So we need to find what information we actually need.

3. The Important Question

Instead of asking:

Which pair of days should I try?

Ask:

Key Question

If today is my selling day, what is the cheapest price I could have bought at before today?

For:

$$[7, 1, 5, 3, 6, 4]$$

when we reach 5:

cheapest previous price = 1

Therefore:

$$profit = 5 - 1 = 4$$

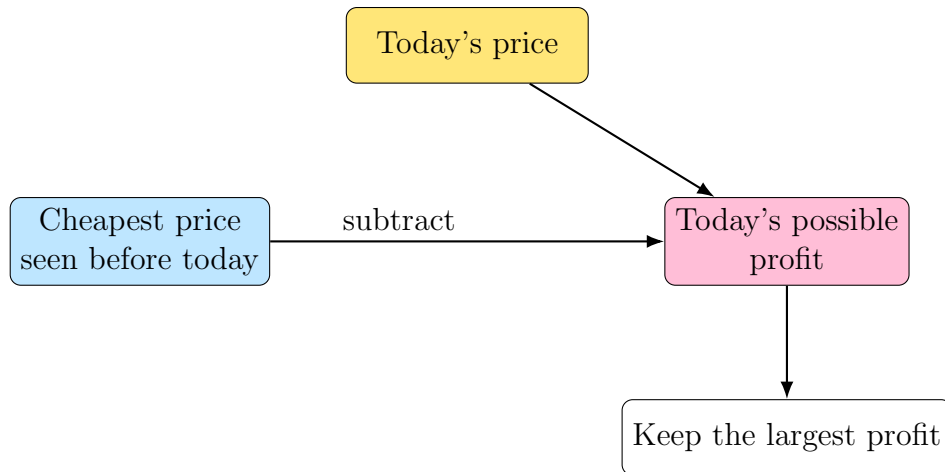
When we reach 6:

$$profit = 6 - 1 = 5$$

So we only need to remember the cheapest price seen so far.

4. Mental Model

Think of walking through the array from left to right.



The flow is:

Price arrives → remember cheapest → calculate today's profit → keep best profit

5. Why Do We Need Two Variables?

We only need to carry two pieces of information:

minimum_price

The cheapest price seen so far.

Best possible buying price

and:

max_profit

The largest profit found so far.

Best transaction found

We do not need to remember every previous price.

Why?

Because for today's selling price, only the **cheapest previous price** can give us the maximum profit.

6. Walk Through the Example

Consider:

[7, 1, 5, 3, 6, 4]

Price = 7

Cheapest price:

$$minimum = 7$$

There is no profit yet:

$$profit = 0$$

Price = 1

We found a cheaper buying price:

$$minimum = 1$$

Price = 5

If we sell today:

$$5 - 1 = 4$$

Therefore:

$$maxProfit = 4$$

Price = 3

$$3 - 1 = 2$$

No improvement.

Price = 6

$$6 - 1 = 5$$

New best:

$$maxProfit = 5$$

Price = 4

$$4 - 1 = 3$$

No improvement.

Final answer:

$$\boxed{5}$$

7. The General Rule

For every price:

1. Update the cheapest price seen so far.
2. Calculate how much profit we could make by selling today.
3. Keep the largest profit.

Conceptually:

$$\textit{minimumPrice} = \min(\textit{minimumPrice}, \textit{price})$$

Then:

$$\textit{profit} = \textit{price} - \textit{minimumPrice}$$

And:

$$\textit{maxProfit} = \max(\textit{maxProfit}, \textit{profit})$$

8. Why This Is Correct

Suppose today is the selling day.

The selling price is fixed:

$$\textit{sell} = \textit{price}$$

To maximize:

$$\textit{profit} = \textit{sell} - \textit{buy}$$

we want the smallest possible:

$$\textit{buy}$$

Therefore:

$$\text{Best profit for today} = \text{today's price} - \text{cheapest previous price}$$

By checking every day as a possible selling day, we cannot miss the optimal transaction.

9. Code

```
class Solution:
    def maxProfit(self, prices):
        minimum_price = prices[0]
        max_profit = 0

        for price in prices:
            minimum_price = min(minimum_price, price)

            profit = price - minimum_price

            max_profit = max(max_profit, profit)

        return max_profit
```

10. Connect the Code to the Mental Model

```
minimum_price = min(minimum_price, price)
```

means:

Remember the cheapest buying opportunity

Then:

```
profit = price - minimum_price
```

means:

What if I sell today?

Finally:

```
max_profit = max(max_profit, profit)
```

means:

Is today's transaction my best one?

11. Complexity

We scan the array only once.

Therefore:

$O(n)$ time

We only store two variables:

minimum_price, max_profit

Therefore:

$O(1)$ space

MEMORY MODEL

Imagine walking through the prices with two pockets:

Pocket 1: Cheapest price I've seen

Pocket 2: Best profit I've made

Every new price asks:

“If I sell today, how much can I make?”

THE ONE THING TO REMEMBER

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Keep the cheapest price so far, and use every new price as a possible selling price.

You do not need to remember every pair of days. You only need the **best buying opportunity so far** and the **best profit so far**.